Task 5: Exploratory Data Analysis (EDA)

Titanic Passenger Dataset – Findings Report

Objective: Explore passenger characteristics and survival outcomes using Pandas, Matplotlib and Seaborn. This report summarizes the commonly used Kaggle Titanic training file (891 passenger records).

1. Dataset overview

The standard training dataset contains 891 rows and 12 columns. The target variable is **Survived** (0 = did not survive, 1 = survived). Important fields include passenger class, sex, age, ticket fare, family counts and embarkation port.

Measure	Finding
Records	891
Survived	342 (38.4%)
Did not survive	549 (61.6%)
Missing Age	177
Missing Cabin	687
Missing Embarked	2

2. Descriptive statistics and data quality

The age distribution is concentrated around young adulthood (mean age approximately 29.7 years; median approximately 28 years). Fare is right-skewed: most passengers paid relatively modest fares, while a small number paid much higher amounts. Cabin has substantial missingness and should not be filled with a single arbitrary value. For modeling or further analysis, missing Age can be imputed with the median and Embarked with the mode, while retaining an indicator for missingness if appropriate.

3. Main visual and statistical insights

• **Survival distribution:** 342 of 891 passengers survived (about 38.4%), so the outcome classes are imbalanced.

• **Sex:** Women had a much higher survival rate (about 74.2%) than men (about 18.9%).

• **Passenger class:** Survival was highest in first class (about 63%), followed by second class (about 47%) and third class (about 24%).

• **Age:** Survivors tended to include more children; however, age alone does not explain survival.

• **Fare:** Fare values show high-value outliers and are generally higher in first class. Fare and survival have a positive association in this sample.

• **Relationships:** Pclass and Fare are associated. The numeric correlation heatmap is useful for a first pass, but it cannot represent categorical Sex unless it is encoded.

4. Recommended visualizations and observations

Visualization	What it reveals
Histogram (Age)	Distribution of passenger ages; concentration among young adults.
Count plot (Survived)	More passengers did not survive than survived.

Bar plot (Sex / Pclass vs Survived)	Large survival-rate differences by sex and passenger class.
Boxplot (Age / Fare)	Spread, skewness and potential outliers.
Pairplot	Pairwise patterns among numeric variables.
Correlation heatmap	Strength and direction of linear relationships among numeric variables.

5. Conclusion

The EDA indicates that survival was strongly associated with sex and passenger class in the Titanic training sample. Age and fare show skewness and potential outliers, while Cabin and Age contain missing data that require careful handling. These findings are descriptive and do not establish causation. The accompanying Jupyter Notebook contains executable Pandas and Seaborn code to reproduce the analysis and generate the visualizations.

Note: These summary figures refer to the standard Kaggle Titanic train.csv dataset (891 rows). If a different Titanic dataset/version is used, the exact counts and percentages may differ.